

Programs on Structures and Pointers:-

9.a Write a C Program to read marks of three students in 3 subjects. Calculate the total marks scored, student wise and subject wise using structure.

```
#include <stdio.h>
#define NUM_STUDENTS 3
#define NUM_SUBJECTS 3

struct Student {
    int marks[NUM_SUBJECTS];
    int totalMarks;
};

int main() {
    struct Student students[NUM_STUDENTS] = {{0}, {0}, {0}};
    for (int i = 0; i < NUM_STUDENTS; i++)
    {
        printf("\n\n*****ENTER MARKS STUDENT %d*****:\n\n", i
+ 1);
        for (int j = 0; j < NUM_SUBJECTS; j++)
        {
            printf("\tEnter marks for Subject %d: ", j + 1);
            scanf("%d", &students[i].marks[j]);
            students[i].totalMarks += students[i].marks[j];
        }
        printf("\nTOTAL MARKS SCORED BY EACH
STUDENT:\n\n");
        for (int i = 0; i < NUM_STUDENTS; i++)
        {
            printf("\t\nStudent %d Total Marks: %d\n", i + 1,
students[i].totalMarks);
        }
        // Calculate subject-wise total marks
        int subjectTotal[NUM_SUBJECTS] = {0};
        for (int i = 0; i < NUM_STUDENTS; i++)
        {
            for (int j = 0; j < NUM_SUBJECTS; j++)
            {
                subjectTotal[j] += students[i].marks[j];
            }
        }
        // Print total marks scored in each subject
```

```

printf("\n\n\tTotal marks scored in each subject:\n\n");
for (int j = 0; j < NUM_SUBJECTS; j++)
{
printf("Subject %d Total Marks: %d\n", j + 1, subjectTotal[j]);
}
return 0;
}

```

Output:-

*****ENTER MARKS STUDENT 1*****:

Enter marks for Subject 1: 55

Enter marks for Subject 2: 66

Enter marks for Subject 3: 54

*****ENTER MARKS STUDENT 2*****:

Enter marks for Subject 1: 33

Enter marks for Subject 2: 44

Enter marks for Subject 3: 45

*****ENTER MARKS STUDENT 3*****:

Enter marks for Subject 1: 44

Enter marks for Subject 2: 23

Enter marks for Subject 3: 33

TOTAL MARKS SCORED BY EACH STUDENT:

Student 1 Total Marks: 175

Student 2 Total Marks: 122

Student 3 Total Marks: 100

Total marks scored in each subject:

Subject 1 Total Marks: 132

Subject 2 Total Marks: 133

Subject 3 Total Marks: 132

9 b. Write a C Program to swap two numbers using pointers and display the result.

```

#include<stdio.h>

void swap(int *x,int *y)
{
    int t;
    t = *x;
    *x = *y;
    *y = t;
}

int main()
{
    int num1,num2;

    printf("Enter value of num1: ");
    scanf("%d",&num1);
    printf("Enter value of num2: ");
    scanf("%d",&num2);

    //displaying numbers before swapping
    printf("Before Swapping: num1 is: %d, num2 is: %d\n",num1,num2);

    //calling the user defined function swap()
    swap(&num1,&num2);

    //displaying numbers after swapping
    printf("After Swapping: num1 is: %d, num2 is: %d\n",num1,num2);

    return 0;
}

```

Output:-

Enter the value of num1:32

Enter the value of num2:24

Before Swapping num1 is 32 and num2 is 24.

After Swapping num1 is 24 and num2 is 32

9.c. Write a C Program to read an array of elements, Compute its sum using pointers.

```

#include <stdio.h>
#include <malloc.h>
void main()
{
int i, n, sum = 0;
int *a;

printf("Enter the size of array A \n");
scanf("%d", &n);

a = (int *) malloc(n * sizeof(int));

printf("Enter Elements of the List \n");
for (i = 0; i < n; i++)
{
scanf("%d", a + i);
}

/* Compute the sum of all elements in the given
array */

for (i = 0; i < n; i++)
{
sum = sum + *(a + i);
/* this *(a+i) is used to access the value stored at the
address*/
}

printf("Sum of all elements in array = %d\n", sum);
return 0;
}

```

Output:-

Enter the size of array A

20

Enter Elements of the List

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

21 Sum of all elements in array = 210